الغيوم في حوسبة القوم

Lecture 1 — The Era of Cloud Computing

> **Cloud Computing** : renting computing (servers, storage, networks, software) from a provider over the internet, on demand, instead of buying and running your own machines.

> **Bare-metal era** : one organization buys, owns, cools, patches and physically secures its own servers in its own data room.

Q/ What were the problems of the bare-metal era ?

1. High **CAPEX**: hardware is bought up front, before demand is known.
2. **Fixed capacity**: you either over-provision (waste) or under-provision (outage).
3. Every patch, backup and physical control is the owner's job alone.
4. Scaling means buying, shipping and racking new machines: **weeks, not minutes**.

> **The Power Wall** : around the mid-2000s chips could no longer get meaningfully faster, because higher clock speeds mean exponentially more **heat and power** draw.

Q/ What was the industry's answer to the Power Wall ?

- Stop chasing raw speed: add **more cores** and **more machines**.
- Spread the work across them: **parallel and distributed computing**.
- This is why cloud data centers are built from **thousands of ordinary machines**, not one extraordinary one.

Q/ What is "scaling out" (horizontal scaling) and what are its building blocks ?

1. **Server**: one physical machine (CPU, RAM, disk, NIC).
2. **Rack**: 10–40+ servers in one cabinet, sharing power and networking.
3. **Aisle / Pod**: rows of racks grouped for airflow, power and network design.
4. **Data center**: thousands of racks, the physical body of "the cloud".

> **Elastic Computing** : the ability to automatically **acquire** more computing resources when demand rises and **release** them when demand falls, paying only for what is actually used.

Example: an exam-result portal scales up for the result-day spike, then scales back down: no manual purchase and no idle hardware afterward.

> **Multi-tenant cloud** : many independent customers (**tenants**) share the same physical hardware, while each tenant's data and workloads stay **logically isolated**.

Q/ Who enforces isolation in a multi-tenant cloud ?

- **Software**: the hypervisor or the container runtime. Its correctness is now a **core security control**.

Q/ What new attack surface does multi-tenancy create ?

1. **Hypervisor escape** (breaking out of a VM).
2. **Side-channel attacks** (e.g. cache timing between neighbours).
3. **Noisy-neighbour** resource abuse.

None of these exist in a single-tenant bare-metal setup.

Service models

> **IaaS (Infrastructure as a Service)** : the provider supplies raw virtualized compute, storage and networking; you install and manage everything above it.

> **PaaS (Platform as a Service)** : the provider manages the OS, runtime and middleware; you deploy only your code and data.

> **SaaS (Software as a Service)** : the provider runs the whole application; you just use it (usually in a browser) and manage your data and user access.

> **DaaS (Desktop as a Service)** : a complete virtual desktop (OS, apps, user profile) streamed from the cloud to any device. It is really **SaaS applied to the desktop**.

	IaaS	**PaaS**	**SaaS**	**DaaS**
You manage	OS, runtime, apps, data	Apps & data only	Your data & access	Your files & profile
Provider manages	Virtualization, servers, storage, network	+ OS, runtime, middleware	The entire stack	Full OS + app stack

	IaaS	PaaS	SaaS	DaaS
Examples	Amazon EC2, Azure VMs, Google Compute Engine	Azure App Service, Google App Engine, Heroku	Google Workspace, Microsoft 365, Salesforce, Zoom	Amazon WorkSpaces, Windows 365, Citrix DaaS
Typical users	Sysadmins, DevOps, infrastructure engineers	Application developers, small teams	Everyone	Remote workforces, call centers, students
Control	Maximum	Medium	Minimum	Minimum

Rule: moving **left → right**, the provider takes more of the **management burden** and more of the **security responsibility**.

Q/ In SaaS, what does the customer still own ?

1. Identity and access management (account security).
2. Data classification and the data you upload.
3. End-user device security.

Q/ Why do organizations use DaaS ?

- Consistent, centrally-managed desktops for remote or contract staff.
- Sensitive data **never leaves the data center**: the endpoint only displays pixels, so a lost laptop does not mean lost data.

Why it matters for a security engineer

> **Shared Responsibility** : every cloud service has a line between what the **provider** secures and what **you** secure. The line **moves with the service model**.

Q/ What are the three security takeaways of Lecture 1 ?

1. **Shared responsibility**: know exactly where the line sits before designing a control.
2. **New attack surface**: multi-tenancy, hypervisors and APIs did not exist in the bare-metal era.
3. **Elasticity cuts both ways**: auto-scaling defends against load spikes, but a misconfigured resource can also scale an attacker's **cost abuse or cryptomining**.

Q/ Bare metal vs cloud — compare.

Point	Bare metal	Cloud
Cost	CAPEX, paid up front	Pay for what you use
Capacity	Fixed at purchase	Elastic, on demand
Time to scale	Weeks	Minutes
Hardware	Owned by one organization	Shared by many tenants
Security duty	All on the owner	Shared with the provider

One-sentence summary: the Power Wall forced computing to scale horizontally, giving rise to elastic, multi-tenant clouds offered at four levels — IaaS, PaaS, SaaS, DaaS — each shifting the security boundary.

Q/ (Discussion) A bank moves its core app from bare metal to SaaS. What controls does it lose direct control over ?

- OS and patching, the runtime, the application code, physical security and network controls: all move to the provider. The bank keeps identity/access, its data, and its users' devices.

Summary of Cloud Computing L1: The Era of Cloud Computing. Built only from the lecture file; check the original slides for anything you're unsure about.